accepted model of DNA is the double helical structure of **James Watson** and **Francis Crick**. They proposed the **three-dimensional model of DNA** on the basis of X-ray diffraction studies of DNA obtained by Rosalind Franklin and Maurice Wilkins. In appreciation of their discoveries on the molecular structure of nucleic acids Watson, Crick and Wilkins were awarded Nobel prize for Medicine in 1962.

QXW5PI

Chemical Composition of DNA molecule

DNA is a large molecule consisting of millions of nucleotides. Hence, it is also called a **polynucleotide**. Each nucleotide consists of three components.

1. A sugar molecules – Deoxyribose sugar.
2. A nitrogenous base.
   There are two types of nitrogenous bases in DNA. They are
   (a) Purines (Adenine and Guanine)
   (b) Pyrimidines (Cytosine and Thymine)
3. A phosphate group

Nucleoside and Nucleotide

Nucleoside = Nitrogen base + Sugar

Nucleotide = Nucleoside + Phosphate

The nucleotides are formed according to the purines and pyrimidines present in them.

18.6.1 Watson and Crick model of DNA

1. DNA molecule consists of two **polynucleotide** chains.
2. These chains form a **double helix** structure with two strands which run **anti-parallel** to one another.
3. **Nitrogenous bases** in the centre are linked to **sugar-phosphate** units which form the backbone of the DNA.
4. Pairing between the nitrogenous bases is very specific and is always between purine and pyrimidine linked by hydrogen bonds.
   * Adenine (A) links Thymine (T) with two hydrogen bonds (A = T)
   * Cytosine (C) links Guanine (G) with three hydrogen bonds(C ≡ G)

This is called **complementary base pairing**.



Figure 18.6 Structure of DNA

5. Hydrogen bonds between the nitrogenous bases make the DNA molecule stable.
6. Each turn of the double helix is 34 A° (3.4 nm). There are ten base pairs in a complete turn.
7. The nucleotides in a helix are joined together by phosphodiester bonds.

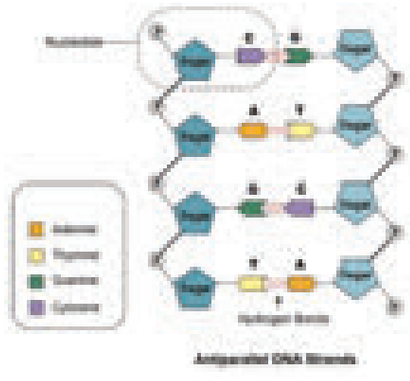


Figure 18.7 Nucleotides in a DNA

More to Know

Chargaff rule of DNA base pairing

Erwin Chargaff states that in DNA, the proportion of adenine is always equal to that of thymine. and the proportion of guanine always equal to that of cytosine.

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